

Solução do Problema Boeing1 - Solver HiGHS

Análise Computacional

6 de outubro de 2025

1 Informações do Problema

Informações do Problema:

- Nome: Boeing1
- Número de Variáveis: 384
- Número de Restrições: 351
- Inviabilidade Primal: 0.000e+00
- Inviabilidade Dual: 2.776e-17
- Valor Primal: -3.352e+02
- Valor Dual: -3.268e+02
- Gap: -2.526e-19
- Número de Iterações: 24

2 Variáveis Primais e Custos Reduzidos ($x \geq 0$ e $z \geq 0$)

Tabela 1: Variáveis primais (x) e custos reduzidos (z)
com valores ≥ 0 do problema Boeing1

Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
2	1.200e+01	3	5.516e-02
5	1.400e+01	4	6.879e-02
10	4.900e+01	6	1.330e-02
11	7.300e+01	7	1.331e-02
12	6.000e+01	9	2.994e-02
13	1.000e+00	14	8.162e-03
15	3.000e+00	16	8.152e-03
17	4.717e+01	22	1.629e-02
18	3.108e+02	23	6.858e-03
19	3.250e-01	25	5.219e-03
20	3.177e+02	33	2.301e-03
24	2.600e+01	39	5.886e-02

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
26	3.700e+01	41	5.887e-02
29	2.150e+02	47	8.073e-04
31	2.700e+01	48	8.173e-04
32	8.300e+01	59	8.266e-03
34	2.780e+02	61	8.266e-03
35	1.040e+02	66	4.499e-03
36	3.304e+01	77	9.982e-03
37	6.821e+01	81	2.565e-02
38	9.800e+01	82	1.566e-02
40	5.000e+00	83	1.566e-02
42	1.000e+00	84	4.747e-03
44	1.578e+02	92	1.284e-01
46	3.500e+01	103	2.793e-03
49	9.421e+01	106	1.284e-01
51	3.979e+01	107	1.000e-05
52	7.001e+01	116	1.868e-01
53	3.430e+02	121	5.483e-03
55	3.700e+01	125	5.483e-03
56	3.856e+01	128	9.797e-04
57	6.956e+00	129	3.773e-03
58	3.249e+01	131	8.947e-04
62	5.460e+02	137	5.483e-03
63	3.721e+00	141	5.483e-03
64	1.390e+02	156	5.219e-03
65	6.627e+01	158	5.272e-03
67	7.500e+01	173	5.308e-05
68	5.416e+00	176	1.108e-02
69	3.304e+01	178	9.742e-03
70	2.608e+02	180	1.106e-02
71	8.400e+02	186	5.242e-03
72	8.791e+01	188	6.732e-03
73	1.680e+02	189	1.000e-05
74	9.800e+01	198	2.764e-02
75	1.464e+03	199	4.310e-02
76	8.581e+00	201	2.763e-02
79	2.242e+01	202	4.310e-02
80	4.100e+01	223	5.903e-01
85	1.538e+02	225	2.794e-01
86	3.924e+01	227	2.766e-01
89	2.000e+00	228	7.064e-02
91	1.510e+02	229	1.636e-01
94	5.100e+01	231	1.630e-01
96	1.300e+01	233	1.323e-01
97	3.789e+01	234	3.731e-02
98	7.715e+01	235	9.319e+00
99	6.956e+00	236	4.804e+00
100	1.770e+02	237	2.631e+00
101	2.310e+01	238	7.588e+00
102	2.490e+01	239	2.477e+00

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
105	6.800e+01	241	1.139e-01
108	2.400e+01	245	2.840e-02
110	2.910e+02	246	9.466e-02
111	6.700e+01	250	4.909e+00
113	8.400e+02	251	2.931e+00
114	1.467e+02	252	2.106e+00
115	2.366e+02	253	1.034e+00
117	5.076e+01	254	5.225e-01
118	1.680e+02	255	2.609e-01
122	8.342e+01	256	8.321e-02
123	7.580e+01	257	7.713e-02
126	7.920e+01	259	1.951e-04
127	1.649e+02	261	2.776e-03
130	3.920e+02	262	6.165e-02
132	7.001e+01	264	1.270e+00
134	5.076e+01	265	6.734e-01
135	4.717e+01	266	3.687e-01
139	6.000e+00	267	7.630e-02
143	1.700e+01	268	1.496e-01
144	2.608e+02	270	2.274e-03
145	3.715e+01	273	6.275e-02
146	1.940e+01	275	6.247e-02
148	9.160e+01	277	6.356e-03
150	5.700e+01	278	2.225e-02
151	9.100e+01	279	2.234e-02
152	3.641e+01	281	8.320e-02
153	1.341e+02	282	4.219e-03
154	1.397e+02	283	3.377e-03
155	4.767e+00	285	5.828e-02
157	2.343e+02	286	1.090e-02
159	1.787e+02	289	1.172e-01
160	1.903e+01	293	1.024e-01
161	2.667e+02	294	7.011e-03
163	4.129e+02	297	1.572e-01
164	1.913e+03	298	1.004e-02
165	1.468e+01	299	3.029e-02
166	2.713e+02	301	9.627e-01
167	1.380e+02	302	4.968e-01
168	4.352e+00	303	2.586e-01
169	2.413e+02	304	7.154e-02
170	1.264e+02	305	8.967e-02
171	1.468e+01	307	6.309e-01
172	1.380e+02	308	2.972e-01
174	1.253e+02	309	1.327e-01
175	1.100e+01	313	3.055e-04
177	7.000e+00	314	2.520e-03
179	4.000e+00	315	3.037e-02
181	2.600e+01	317	4.127e-02
182	2.000e+00	319	4.992e-03

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
183	1.000e+00	320	1.782e-02
184	1.100e+01	321	2.414e-02
187	2.400e+01	324	2.839e+01
190	9.706e+00	330	1.312e+00
191	6.294e+00	331	6.225e-01
192	3.000e+00	332	3.266e-01
193	2.400e+01	334	3.221e-02
194	4.000e+01	337	5.950e-02
195	5.000e+00	338	2.638e-02
196	1.000e+00	340	5.835e+00
197	1.300e+01	341	1.900e+00
200	8.000e+00	344	3.333e-03
203	1.000e+00	345	6.150e-02
204	1.000e+00	349	2.014e-01
205	1.000e+00	350	3.697e-01
206	1.000e+00	351	5.824e-02
207	2.000e+00	355	1.226e-01
208	7.000e+00	357	1.002e-01
209	2.000e+00	358	1.663e-01
210	2.000e+00	361	1.615e-02
211	2.000e+00	363	1.002e-01
212	1.000e+00	364	1.501e-01
213	2.000e+00	365	4.360e-02
214	5.000e+00	367	4.723e-01
215	2.000e+00	368	7.870e-01
216	2.100e+01	369	6.730e+00
217	1.300e+01	370	2.186e+00
218	6.000e+00	373	6.528e+00
219	7.000e+00	374	2.084e+00
220	1.700e+01	376	5.254e-03
221	2.500e+01	377	5.460e-02
222	6.294e+00	379	3.472e-01
240	2.355e+00	380	1.612e-01
242	1.603e-01	381	6.515e-02
243	7.214e-01	382	4.829e-05
244	2.800e-01		
247	8.594e-02		
248	8.382e-01		
249	7.583e-02		
258	4.658e+00		
260	7.488e-02		
263	1.922e+00		
269	7.000e+00		
271	3.085e+00		
272	6.952e-01		
274	3.897e+00		
276	1.511e+00		
280	2.417e+00		
284	9.242e-01		

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
287	1.691e+00		
288	2.228e-01		
290	7.048e-01		
291	1.595e-01		
292	2.136e+00		
295	7.000e-01		
296	3.300e+00		
300	1.618e-01		
306	1.196e+00		
310	2.345e-01		
311	7.655e-01		
312	2.228e-01		
316	3.201e+00		
318	6.642e-01		
322	2.454e+00		
325	1.000e+00		
326	4.767e-01		
328	5.233e-01		
329	1.000e+00		
333	8.275e-01		
335	4.860e-01		
336	5.005e-01		
339	2.062e+00		
342	3.175e+00		
343	6.828e-01		
346	7.269e-01		
347	2.530e-01		
348	1.083e-01		
353	2.384e-01		
354	1.271e+00		
356	2.078e+00		
359	9.513e+00		
360	8.857e+00		
362	2.000e+01		
366	1.725e-01		
371	1.000e+00		
372	1.000e+00		
375	1.000e+00		
378	7.319e-02		
383	4.652e+00		
384	2.000e+01		

3 Variáveis Duais (Multiplicadores de Lagrange - Valores λ_i)

Tabela 2: Variáveis duais (y) com valores ≥ 0 do problema Boeing1

Coordenada y	Valor y (Dual)
3	3.996e-01
13	3.859e-02
15	8.637e-02
19	5.483e-03
21	4.758e-02
22	1.330e-02
23	5.127e-02
25	5.732e-02
27	5.208e-02
29	5.881e-02
30	4.552e-02
31	1.986e-02
32	6.463e-03
33	2.261e-02
34	1.330e-02
35	9.256e-03
36	2.764e-02
37	5.483e-03
38	2.261e-02
39	1.330e-02
40	2.479e-02
41	2.910e-02
43	2.910e-02
45	2.261e-02
46	8.964e-03
47	2.261e-02
48	8.791e-03
49	2.764e-02
51	6.378e-03
52	1.936e-02
53	5.483e-03
54	2.634e-02
55	1.880e-02
56	2.344e-02
58	1.339e-01
60	5.483e-03
66	1.198e-02
69	2.261e-02
72	9.534e-03
73	2.261e-02
74	5.483e-03
77	5.035e-03
78	2.261e-02
81	5.483e-03
82	2.577e-02
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Coordenada y	Valor y (Dual)
83	2.454e-02
84	1.070e-02
85	8.970e-03
86	2.155e-02
87	3.002e-02
88	8.970e-03
90	1.070e-02
91	2.577e-02
92	1.890e-02
93	4.471e-02
94	8.970e-03
95	2.460e-02
96	1.070e-02
97	2.068e-02
98	3.002e-02
99	1.070e-02
100	8.970e-03
101	1.070e-02
102	8.970e-03
103	4.552e-02
107	5.208e-02
108	1.330e-02
110	2.764e-02
111	6.722e-03
112	1.330e-02
209	1.011e-01
210	5.499e-01
211	7.713e+00
218	2.041e-01
219	1.568e+00
224	2.538e-01
225	5.749e-02
228	3.315e+01
233	7.267e-03
234	1.134e-01
235	2.634e-01
251	8.761e-02
252	9.153e+00
253	1.507e+00
254	2.969e+01
265	4.496e-03
266	8.073e-04
269	7.500e-02
270	7.500e-02
271	5.515e-02
272	7.500e-02
273	8.049e-02
275	2.894e-03
282	2.853e-03

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Coordenada y	Valor y (Dual)
284	2.764e-02
285	1.566e-02
292	3.767e-03
293	8.266e-03
296	1.566e-02
300	7.264e-04
304	1.455e-03
306	2.474e-03
310	5.665e-04
315	1.868e-01
328	9.797e-04
329	3.312e-03
330	1.275e-02
331	5.362e-02
332	1.961e-02
343	4.218e-03
346	5.031e-02
347	5.642e-03
349	5.642e-03
351	3.862e-03